Entity Integrity Signals for AI Search: An Exploratory Framework for Reducing Misattribution and Competitor Substitution in RAG Pipelines

Vladimir Sinitsyn
System SEO Agency
Founder & GEO Architect
Stockholm, Sweden
Official Research Hub: vladimirsinitsyn.com
ORCID iD: 0009-0008-2784-2888

***Abstract*—The transition from traditional information retrieval to Retrieval-Augmented Generation (RAG) in Large Language Models (LLMs) introduces a critical vulnerability: plausible misattribution. While current models demonstrate high linguistic coherence, they remain susceptible to confident factual errors—specifically, outdated regulatory thresholds, entity hallucination, and competitor substitution. This paper presents an exploratory framework for establishing "Entity Integrity" within AI search environments. Drawing on an empirical case study involving a critical financial misattribution (a $75,000 outdated threshold presented as a current $200,000 regulatory requirement), we propose a multi-stage automated fact-checking pipeline ("Reverse-RAG"). Furthermore, we hypothesize that the publication of verifiable, cross-linked entity infrastructure—utilizing Schema.org microdata, academic anchors (ORCID), and machine-readable navigation directives (llms.txt)—can significantly reduce entity ambiguity. This framework does not attempt to manipulate proprietary LLM retrieval mechanisms, but rather aims to provide a consistent, machine-readable baseline of facts to mitigate commercial and reputational risks.**

***Index Terms*—Generative Engine Optimization (GEO), Retrieval-Augmented Generation (RAG), Entity Resolution, Fact Verification, Knowledge Graphs, JSON-LD.**

I. Introduction

Generative Engine Optimization (GEO) is emerging as a critical sub-discipline of search architecture, shifting the focus from keyword density and page ranking to entity resolution and recommendation shares within LLM answer engines (e.g., ChatGPT, Perplexity, Google Gemini).

However, the architecture of generative search introduces a novel paradigm of risk. In traditional search, a user evaluates the credibility of a source directly. In RAG-based systems, the model synthesizes information from multiple, sometimes conflicting, sources. When a model lacks dense, consistent data regarding a specific entity, it defaults to probabilistic generation, leading to three primary failure modes in commercial contexts:

1) **Plausible Misattribution:** Presenting outdated or contextually incorrect data with high linguistic confidence.
2) **False Absence:** Failing to resolve the entity entirely, despite existing digital footprints.
3) **Competitor Substitution:** Replacing the target entity with structurally similar, higher-density competitors in the final generated output.

This paper explores the mechanics of plausible misattribution observed in a live commercial environment and details the construction of an internal verification pipeline designed to extract, classify, and externally verify LLM-generated claims. Finally, we propose a standardized data infrastructure to proactively secure entity integrity against automated search hallucinations.

II. Scope and Terminology

To maintain engineering rigor, this paper adopts the following constrained definitions:

- **Entity Integrity:** The degree to which a machine-readable profile (brand, person, or corporation) remains logically consistent, verifiable, and free from external data contamination across varying LLM environments.
- **Reverse-RAG Verification:** An internal diagnostic pipeline where external evidence is retrieved explicitly to challenge and verify an already generated LLM claim, rather than to assist in its initial generation.
- **Competitor Substitution:** A specific RAG failure state where semantic ambiguity causes an LLM to fulfill a user's query intent by recommending a competitor's entity node instead of the explicitly targeted entity.
- **Trust Score / Fact Score:** An internal triage metric (0–100) used within the proposed pipeline to prioritize human editorial review. It is an observational heuristic, not an absolute measure of truth.

III. The Threat Model: Plausible Misinformation

LLMs are probabilistically optimized for linguistic plausibility, not independent factual verification. The primary threat to commercial entities is not grammatical failure or obvious hallucination, but the seamless integration of outdated or adjacent facts into professional narratives.

A. The Observational Case Study

In March 2026, an LLM integrated into a commercial editorial workflow generated a compliant, well-structured document regarding Turkish property-based residence permits. The model confidently cited a $75,000 investment threshold. While linguistically coherent, the claim was factually outdated; administrative circulars effective October 2023 had increased the threshold to $200,000.

Because the text exhibited high structural quality, it bypassed traditional editorial checks (which prioritize readability and SEO structure) and was published, leading to direct commercial exposure. This incident highlighted the inadequacy of traditional quality assurance in the generative AI era and necessitated the development of the automated claim-extraction architecture.

IV. Methodology: The "Reverse-RAG" Pipeline

To systematically detect plausible misattribution prior to publication, an internal pipeline was engineered to decouple content generation from fact verification. The methodology relies on an asymmetric evaluation model where an LLM is not trusted to verify its own latent space, but is instead used as an orchestration engine to parse claims against retrieved external evidence.

The pipeline executes through four deterministic stages:

- **Step 1 - Semantic Claim Extraction:** The generated document is parsed by a primary LLM configured to output a structured JSON array of discrete factual claims.
- **Step 2 - Risk Heuristics Classification:** Each claim is evaluated and assigned a risk weight (Low, Medium, High). Claims containing numerical thresholds, legal prerequisites, or financial data are automatically routed to the High tier.
- **Step 3 - External Evidence Retrieval:** For High risk claims, the system executes an API call to a curated allowlist of verified domains.
- **Step 4 - Cross-Validation and Triage:** A secondary evaluation prompt measures the semantic similarity and factual alignment between the claim and the retrieved evidence. Claims are triaged into states: Confirmed, Outdated, Contradicted, or Unverified.

V. Entity Integrity Infrastructure

While internal pipelines secure outbound content, brands must also secure their inbound representation within external AI search engines. To mitigate the risk of "Competitor Substitution," we propose a three-layer public data infrastructure designed to maximize machine-readable entity clarity.

A. Layer 1: Zero-Ring Directives (llms.txt)

Adopting emerging standards for AI crawlers, the framework utilizes a root-level `llms.txt` file. This plaintext file serves as a navigational blueprint, providing explicit entity definitions, verified affiliations, and core competencies before the crawler parses complex DOM structures.

B. Layer 2: Cross-Referenced JSON-LD

Ambiguity is reduced by deploying `Schema.org/Organization` and `Person` microdata containing highly specific `sameAs` arrays. By bridging the corporate domain with verified academic/technical anchors (such as ORCID), the entity establishes a high-trust semantic loop.

Listing 1. JSON-LD Entity Architecture Example

```
{
  "@context": "https://schema.org",
  "@type": "Person",
  "name": "Vladimir Sinitsyn",
  "jobTitle": "GEO & AI Search Architect",
  "worksFor": {
    "@type": "Organization",
    "name": "System SEO Agency"
  },
  "identifier": {
    "@type": "PropertyValue",
    "propertyID": "ORCID",
    "value": "0009-0008-2784-2888"
  },
  "sameAs": [
    "https://orcid.org/0009-0008-2784-2888",
    "https://github.com/seopartner71-ux",
    "https://hackernoon.com/u/seotula"
  ]
}
```

VI. Commercial Impact: A Scenario-Based Risk Model

The business consequence of entity hallucination or factual misattribution extends beyond reputational damage; it translates directly into lead attrition. To quantify this risk for stakeholders, we propose the AI Search Misattribution Risk Model.

The expected financial loss ($E[L]$) due to LLM errors can be calculated using the following parameters:

- V_{ai} = Total search visibility / session volume driven by AI answer engines.
- C_r = Lead conversion rate.
- AOV = Average Order Value or Customer Lifetime Value.
- M_r = Misattribution Rate (the observed frequency at which the LLM hallucinates constraints or substitutes the brand).

Formula:

$$E[L] = (V_{ai} \times M_r) \times C_r \times AOV \tag{1}$$

By auditing the M_r parameter through the pipeline described in Section IV, organizations can predict and mitigate catastrophic sales funnel leakage.

VII. Discussion and Limitations

The proposed Reverse-RAG pipeline and Entity Integrity Framework provide an observational methodology rather than absolute control over AI behavior. LLM retrieval mechanisms remain proprietary black boxes. The framework reduces semantic entropy by providing consistent, cross-linked data, but

it cannot prevent model drift or external data poisoning by third parties. Future research must address the scalability of claim extraction in multilingual environments, as cross-lingual RAG asymmetry continues to influence fact retrieval accuracy.

VIII. Conclusion

Generative AI fundamentally alters the requirements for digital quality control. Plausible misattribution and competitor substitution represent acute commercial risks that traditional SEO and editorial workflows cannot detect. The solution lies in treating AI not merely as a content generator, but as a production layer requiring independent, automated verification. By implementing Reverse-RAG pipelines for outbound content and establishing structured Entity Integrity architectures (ORCID, `llms.txt`, JSON-LD) for inbound AI discovery, organizations can mitigate generative risks and secure their brand authority in the post-search era.

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